

1. Draw the OSI model on paper or using any drawing tool, label all 7 layers, and write one real-world example protocol (like HTTP, TCP, IP, etc.) next to each layer.

<u>OSI MODEL</u>		
<u>LAYER</u>	<u>LAYER NAME</u>	<u>EXAMPLE PROTOCOL</u>
7	Application Layer Provides network services to end-user applications.	HTTP (HyperText Transfer Protocol)
6	Presentation Layer Handles data formatting, encryption, and compression.	TLS (Transport Layer Security)
5	Session Layer Manages sessions and connections between applications.	NetBIOS (Network Basic Input/Output System)
4	Transport Layer Provides reliable data transfer and error recovery.	TCP (Transmission Control Protocol)
3	Network Layer Handles logical addressing and routing of data.	IP (Internet Protocol)
2	Data Link Layer Provides node-to-node data transfer and error detection.	Ethernet (IEEE 802.3)
1	Physical Layer Transmits raw bits over the physical medium.	Physical Medium (e.g., Cables, Hubs, Connectors)

2. Create a table listing 5 popular apps you use (such as WhatsApp, Instagram, Zomato, Flipkart, Paytm) and for each, identify two network protocols that are likely used when you interact with the app (for example, HTTPS, TCP, UDP, DNS, etc.).

No	Popular App	Protocol 1	Protocol 2	Purpose
1	WhatsApp	HTTPS	UDP	HTTPS secures message and account communication whereas UDP is used for voice and video calls.
2	Instagram	HTTPS	DNS	HTTPS securely transfers posts, messages and account data whereas DNS converts Instagram domain names into its IP addresses.
3	Zomato	HTTPS	TCP	HTTPS secures restaurant searches, orders and payments whereas TCP provides reliable data delivery.
4	Flipkart	HTTPS	DNS	HTTPS protects

				browsing, login, shopping and payment data whereas DNS resolves Flipkart's domain name to its server IP.
5	Paytm	HTTPS	TCP	HTTPS protects sensitive transactions and login information whereas TCP provides reliable communication between the app and servers.

3. Use the command prompt or terminal to run 'ping' and 'tracert' (Windows) or 'traceroute' (Linux/Mac) to www.youtube.com, then explain which OSI layers are involved in these commands and how.

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Command Prompt
(c) Microsoft Corporation. All rights reserved.
C:\Users\rohan>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4829:400::] with 32 bytes of data:
Reply from 2001:4860:4829:400::: time=93ms
Reply from 2001:4860:4829:400::: time=52ms
Reply from 2001:4860:4829:400::: time=46ms
Reply from 2001:4860:4829:400::: time=63ms

Ping statistics for 2001:4860:4829:400:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 46ms, Maximum = 93ms, Average = 63ms

C:\Users\rohan>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4829:400::]
over a maximum of 30 hops:
  0  11 ms  3 ms  4 ms  2401:4900:8fc5:c465:26de:8aff:fe79:8d11
  1  *      *      *      Request timed out.
  2  37 ms  39 ms  35 ms  fc00::82
  3  41 ms  42 ms  43 ms  2404:a800:2a00:103::1
  4  61 ms  57 ms  57 ms  2404:a800::167
  5  55 ms  42 ms  53 ms  2001:4860:4829:400::
  6

Trace complete.
C:\Users\rohan>

```

OSI layers involved in ping and tracert commands

OSI Layer	Name	How ping/tracert use it
1	Application	The command is initiated by the user environment. DNS may also resolve www.youtube.com to an IP address.
2	Presentation	Not significantly involved, ICMP itself does not require application-level encryption or formatting.
3	Session	Not significantly involved because ping and tracert don't establish a normal application session.
4	Transport	TCP/UDP are not used by normal windows ping and tracert uses ICMP-based probing by default on windows.
5	Network	Most important layer. IP addressing and ICMP are used. tracert relies on increasing TTL values to identify each router hop.
6	Data Link	Ethernet/Wi-Fi frames carry the IP/ICMP packets across the local network.
7	Physical	Electrical, radio or optical signals physically transmit the data through Ethernet cables or Wi-Fi.

4. **Write a short scenario describing how a message sent from your device to a friend's device on WhatsApp travels through each OSI layer, mentioning what happens at each step.**

Scenario: Sending a WhatsApp Message

- Suppose I send "Hello, how are you?" to my friend on WhatsApp. The message travels through all seven OSI layers on my phone and then moves back up through the layers on my friend's phone.

1) Application Layer (Layer-1)

- I type message in WhatsApp and press send. WhatsApp provides the application service for sending the message.

2) Presentation Layer (Layer-2)

- The message is formatted and encrypted so that it can be securely transmitted over the network.

3) Session Layer (Layer-3)

- This layer manages the communication session between my WhatsApp application and the remote service.

4) Transport Layer (Layer-4)

- The message is divided into data segments and transport protocols such as TCP or UDP help deliver the data between endpoints.

5) Network Layer (Layer-5)

- The data is placed into IP packets. IP addresses are used to route the packets from my device through routers and networks towards the destination.

6) Data-link Layer (Layer-6)

- The packets are placed into frames and the device uses technologies such as Wi-Fi or mobile network protocols to communicate with the next network device.

7) Physical Layer (Layer-7)

- The frames are converted into electrical, radio or optical signals. In my case, the data may travel as Wi-Fi radio signal from my phone to the router.